

Recorded ERP observation bank /
simulated affective-state process

held-out subject-
condition observations



simulated affective state transition

observation x_t

ARSPI-Net evidence estimator

E

spike
embedding

D

dynamical

T

topological

C

coupling

class-prob.
vector $\mathbf{p}_\theta(x_t)$

Dirichlet belief
update

$$\alpha \leftarrow \alpha + \mathbf{p}_\theta(x_t)$$

belief $b(s)$

Expected-free-energy
(EFE) controller

$$a^* = \arg \min_a \text{EFE}(a)$$

policy action / target-state query a